Palatial Pools Technology Disclosure Document

Confidential

Palatial Pool Services Pty Ltd

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1. Introduction

This document provides a comprehensive technical overview of Palatial Pools' proprietary technology suite designed to automate and optimize commercial pool maintenance. Our solution integrates advanced sensing, control, and communication technologies to ensure pool water quality, operational efficiency, and regulatory compliance. This document is intended for engineering stakeholders, prospective clients, partners, and investors requiring detailed insight into the design, implementation, and operational principles of the technology.

The technology suite is developed and maintained by Palatial Pool Services

Pty Ltd, leveraging best practices in embedded systems, secure communications, and data analytics tailored for commercial aquatic environments. All information is presented in accordance with Australian consumer law and industry standards to ensure accuracy, transparency, and compliance.

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2. System Architecture Overview

The Palatial Pools system is a modular, distributed architecture comprising three primary subsystems:

- **Water Quality Monitoring Subsystem (WQMS)**: Real-time sensing of chemical and physical water parameters.
- **Automated Chemical Dispensing System (ACDS)**: Precision dosing of pool chemicals based on monitored data.
- **Environmental Sensing and Weather Integration (ESWI)**: External environmental data acquisition to optimize chemical management.

All subsystems communicate via a secure, wireless mesh network, coordinated by a central control unit (CCU) with cloud connectivity for remote monitoring and analytics.

High-Level Architecture Diagram

```mermaid
graph TD
    WQMS[Water Quality Monitoring Subsystem]
    ACDS[Automated Chemical Dispensing System]
    ESWI[Environmental Sensing & Weather Integration]
    CCU[Central Control Unit]
    CLOUD[Cloud Analytics & Dashboard]

    WQMS --> CCU
    ACDS --> CCU
    ESWI --> CCU
    CCU --> CLOUD
```

3. Core Technology Components

3.1 Water Quality Monitoring Subsystem

Sensor Array

The WQMS incorporates multiple sensor types to measure critical water quality parameters continuously:

Parameter	Sensor Type	Measurement Range	Accuracy
pH	Ion-selective glass electrode	0 - 14	±0.05 pH units
Free Chlorine (Cl_2)	Amperometric sensor	0 - 10 ppm	±0.1 ppm
Temperature	Thermistor (NTC type)	0°C - 50°C	±0.2°C
Turbidity	Optical nephelometric sensor	0 - 10 NTU	±0.1 NTU
ORP (Oxidation Reduction Potential)	Platinum electrode sensor	-1000 mV to +1000 mV	±5 mV

Sensors communicate via industry-standard protocols such as **I^2C** and **SPI** to embedded microcontrollers within the WQMS node.

Calibration and Self-Diagnostics

- Automated calibration routines are scheduled periodically using standard reference solutions to maintain sensor accuracy over time.
- Self-diagnostic algorithms monitor sensor drift and fault detection, triggering alerts if maintenance is required.

3.2 Automated Chemical Dispensing System

Actuation and Control

The ACDS controls dosing pumps with precision flow control valves actuated by stepper motors. The system supports dosing of multiple chemical agents including chlorine, pH adjusters (acid/base), and algaecides.

- Flow rates are controlled via **pulse-width modulation (PWM)** signals based on dosing volume calculations.
- Feedback loops use flow meters (turbine or ultrasonic) to verify actual dispensed volumes.

Control Algorithms

Dosing decisions are computed using a **Proportional-Integral-Derivative (PID) control algorithm** that dynamically adjusts chemical input based on real-time sensor feedback and target setpoints.

The closed-loop control equation is:

$$ u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} $$

Where:

- $e(t)$ = error between target and measured parameter (e.g., pH)
- K_p, K_i, K_d = tuned PID coefficients

Setpoints are configurable via the central control unit interface, adhering to regulatory standards for pool water quality.

3.3 Environmental Sensing and Weather Integration

To enhance chemical dosing efficiency, the ESWI subsystem monitors ambient conditions:

- **Humidity and Temperature Sensors**: Using calibrated digital sensors (e.g., Sensirion SHT3x series).

- **Rainfall Detection**: Optical rain sensors detect precipitation events to suspend dosing or trigger flushing protocols.
- **Weather Forecast Integration**: Cloud service APIs (e.g., Bureau of Meteorology REST API) provide forecast data to anticipate environmental changes.

This data is used to adjust dosing schedules proactively, reducing chemical waste and optimizing pool conditions.

3.4 Communication and Networking

Wireless Mesh Network

Subsystems communicate via **IEEE 802.15.4** compliant radios operating in the 2.4 GHz ISM band, forming a low-power, self-healing mesh network. This enables:

- Redundancy and reliability in communication paths.
- Dynamic node discovery and routing using the **Zigbee PRO** protocol stack.

Central Control Unit (CCU)

- The CCU hosts a real-time operating system (RTOS) managing network coordination, data aggregation, and user interface.
- Interfaces with cloud services over **TLS 1.3** encrypted channels using **AES-256** cipher suites for secure remote access and data upload.

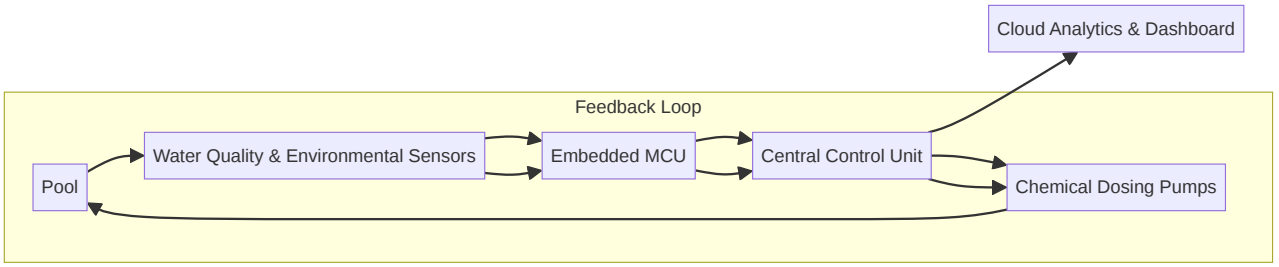
Cloud Connectivity

- Cloud backend hosted in Australian data centers complies with **Australian Privacy Principles (APPs)**.
- Provides dashboard visualization, historical data analytics, and alerting services.

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4. Data Flow and Processing

Data Flow Diagram



Data Processing Pipeline

1. **Data Acquisition**: Sensors sample water parameters every 5 seconds; data is timestamped in UTC.
2. **Preprocessing**: Sensor readings are filtered using a **Kalman Filter** to reduce noise and outliers.
3. **Parameter Estimation**: Chemical concentrations and water quality indices are computed using calibrated sensor models.
4. **Decision Logic**: Control algorithms determine dosing requirements based on measured vs target values.
5. **Command Execution**: Dosing commands sent to actuators with feedback monitoring for validation.
6. **Data Storage and Reporting**: All data logged locally and synchronized with cloud servers every 15 minutes.

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5. Security Architecture

Security is paramount due to the sensitive nature of operational data and remote control capabilities.

Security Aspect	Implementation Detail
Network Security	TLS 1.3 for all cloud communications; Zigbee network uses AES-128 encryption with rotating network keys.
Device Authentication	Mutual authentication using X.509 certificates issued by Palatial Pools' internal PKI.
Data Integrity	Message authentication codes (MAC) applied to all telemetry and command packets.
Access Control	Role-based access control (RBAC) enforced on CCU and cloud portals; multi-factor authentication (MFA) for user accounts.
Firmware Updates	Secure OTA updates with signed firmware images to prevent tampering.
Physical Security	Tamper-evident seals and encrypted storage of sensitive data on embedded devices.

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6. Integration and API Specifications

API Overview

Palatial Pools provides a RESTful API enabling integration with third-party building management systems (BMS) and IoT platforms.

API Endpoint	Method	Description	Data Format
`/api/v1/status`	GET	Retrieves current pool status and sensor data	JSON
`/api/v1/dosing`	POST	Sends dosing commands or setpoint adjustments	JSON
`/api/v1/logs`	GET	Returns historical water quality logs	JSON/CSV
`/api/v1/alerts`	GET	Retrieves active and historical alerts	JSON

Data Formats

Example JSON response for `/api/v1/status`:

```
{
  "timestamp": "2025-04-01T12:00:00Z",
  "waterQuality": {
    "pH": 7.45,
    "freeChlorine": 2.3,
    "temperatureC": 27.1,
    "turbidityNTU": 0.5,
    "ORP_mV": 650
  },
  "environment": {
    "humidityPercent": 55,
    "ambientTempC": 23.0,
    "rainDetected": false
  },
  "systemHealth": {
    "sensorStatus": "OK",
    "pumpStatus": "Operational"
  }
}
```

Integration Methods

- **Authentication**: OAuth 2.0 token-based authentication with scopes for read/write access.
- **Transport Protocol**: HTTPS using TLS 1.3 mandatory; API clients must validate server certificates.
- **Rate Limits**: 1000 requests per hour per client to ensure service reliability.
- **Sandbox Environment**: A dedicated sandbox API endpoint is available for integration testing with simulated data.

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7. Performance and Reliability

- **Uptime SLA**: 99.9% availability guaranteed for cloud services.

- **Sensor Sampling Rate**: Configurable from 1 second to 15 minutes to balance responsiveness and power consumption.
- **Latency**: Command round-trip latency under 500 ms on local network; cloud interaction latency under 2 seconds.
- **Redundancy**: Mesh networking enables alternate routing paths to mitigate node failures.
- **Maintenance Alerts**: Predictive maintenance analytics based on sensor drift patterns reduce unplanned downtime.

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8. Standards Compliance

Palatial Pools technology is designed to comply with the following relevant standards and guidelines:

- **Australian/New Zealand Standard AS/NZS 4276**: Guidelines for pool water quality and safety.
- **ISO/IEC 27001**: Information security management compliance for cloud and data handling.
- **IEEE 802.15.4 & Zigbee PRO**: Wireless communication protocols for sensor networks.
- **TLS 1.3 (RFC 8446)**: Secure communication standards for data in transit.
- **Australian Consumer Law (ACL)**: Ensures no deceptive or misleading claims are made regarding the technology's capabilities.

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9. Information Classification and Handling

Information related to Palatial Pools technology is classified into four levels to ensure appropriate protection and disclosure.

Classification Level	Description	Access & Handling Guidelines
PUBLIC	General information suitable for unrestricted distribution (e.g., marketing materials).	Can be freely shared via public channels.
COMMERCIAL-IN-CONFIDENCE	Detailed technical specifications, pricing, and performance data shared under NDA.	Requires signed NDA; limited recipient list.
PROPRIETARY	Sensitive internal information including source code, algorithms, and operational procedures.	Restricted to internal personnel and contractors under strict confidentiality.
RESTRICTED	Highest security level including encryption keys and strategic plans.	Director-level access only; strict controls.

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10. Disclosure Guidelines

10.1 Sales and Business Development

- Only **PUBLIC** and **COMMERCIAL-IN-CONFIDENCE** information may be disclosed.
- Disclosures under **COMMERCIAL-IN-CONFIDENCE** require a signed NDA.
- Presentations focus on capabilities and business outcomes, never revealing internal algorithms or architectural details.

10.2 Technical Partnerships

- Partners granted access to **COMMERCIAL-IN-CONFIDENCE** technical specifications under mutual NDA.
- API documentation abstracts internal implementation details, describing only interface contracts and data schemas.
- Integration testing conducted within sandbox environments; production system access is restricted.

10.3 Regulatory and Legal

- Regulatory bodies may be granted access to information at any classification level as required by law.
- Legal counsel approval mandatory before disclosing **PROPRIETARY** or **RESTRICTED** information.
- All disclosures are logged with recipient details, date, and scope.

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11. Prohibited Disclosures

The following information must never be externally disclosed:

- Use of third-party AI/ML services; all capabilities presented as proprietary.
- Specific open-source libraries or frameworks underpinning the technology.
- Internal code repository locations or version control details.
- Individual customer data, even anonymized, without explicit consent.
- Security vulnerability information, including details of patches.
- Internal communication tools, workflows, or development methodologies.

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12. Breach Handling and Policy Review

- Suspected or actual breaches must be reported immediately to the Director.
- Incidents will be investigated promptly; outcomes may include disciplinary action, contract termination, or legal proceedings.
- All employees and contractors are required to acknowledge this policy annually.
- The policy undergoes quarterly review to accommodate new technology developments and business changes.
- The current version supersedes all previous versions.

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13. Authorisation

Kieran Goldsmith
Director
Palatial Pool Services Pty Ltd
April 2025

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